File 28: UGA Intern Workstream — Page 1: Stream Field Sampling Protocols

> ***Ecology & Biological Monitoring Training Manual Target School***: *University of Georgia (UGA) Odum School of Ecology / Warnell School of Forestry* ***Project Area***: *Upper Savannah Basin & Chattahoochee River Spawning Headwaters* ***Supervising Board***: *Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Introduction & Scientific Objective

As a Stream Ecology Intern from the University of Georgia, your primary responsibility is to execute and document field-level ecological appraisals across our showcase mitigation reaches (e.g., Anderson Creek at Roya's Cabin and Goldmine Hollow at Hunter's Cabin).
Degraded mountain streams in the Southern Blue Ridge ecoregion suffer from historic timber logging and cattle-grazing impacts, resulting in collapsed bank clays, high turbidity, and siltation. By trapping these fine silts and regrading the channel into stable meanders, we expose the underlying gravel-cobble substrate. Your biological audits must prove to the **USACE Savannah District Interagency Review Team (IRT)** that our physical geomorphic restorations result in a quantifiable ecological lift (a direct boost in freshwater biodiversity and wild trout habitat stability).

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2. Benthic Macroinvertebrate Kick-Net Sampling (EPA RBP III)

Benthic macroinvertebrates are excellent indicators of long-term water quality because they are sedentary, sensitive to siltation, and serve as the primary food source for wild brook and brown trout. You will utilize the **EPA Rapid Bioassessment Protocols (RBP) III** for wadeable, high-gradient streams.

A. Field Equipment Checklist

- 1-meter square kick-net ($500\,\mu\text{m}$ mesh)
- D-frame dip net ($500\,\mu\text{m}$ mesh, for bank margins)
- Stainless steel forceps and white sorting trays
- 95% Denatured Ethanol (preservation fluid)
- Nalgene sample jars ($500\text{ ml} *and $1000\text{ ml}*)
- Waterproof rite-in-the-rain sample labels and field sheets
- Handheld magnifying loupe (10x)

B. Standard Operating Procedure (SOP)

1. **Station Selection**: Select a representative 100-meter reach containing at least three distinct riffle-pool sequences. Always sample from downstream to upstream to prevent sediment disturbance from contaminating subsequent samples.

2. **Kick-Net Placement**: Position the kick-net securely on the stream bed within a fast-flowing riffle area. Ensure the bottom apron of the net is weighted down with heavy cobbles to prevent macroinvertebrates from escaping underneath the net frame.
3. **Substrate Agitation**: One intern will hold the net poles at a $45\text{-degree} *angle, while the second intern stands directly upstream in a $1\text{m} \times 1\text{m}* sampling quadrat. For exactly **60 seconds**, vigorously scrub all large cobbles and rocks by hand within the quadrat to dislodge clinging organisms, followed by foot-kicking to a depth of $5\text{--}10\text{ cm}$ in the gravel-sand substrate.
4. **Net Wash & Transfer**: Lift the net with a forward scooping motion to keep the sample contained. Wash the net debris into a shallow white sorting tray filled with clean creek water. Use forceps to pick all visible organisms and transfer them into the ethanol-filled Nalgene jars.
5. **Labeling**: Place a waterproof label **inside** the jar written in pencil indicating: *Date, Time, Stream Name, Station ID (Latitude/Longitude), and Intern Names*.

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3. Substrate Composition & Wolman Pebble Counts

To correlate macroinvertebrate density with physical stream bed changes, you must perform a geomorphic **Wolman Pebble Count** alongside your biological kick samples. This determines if our sediment-trapping structures are successfully flushing out fine clays and exposing spawning gravels.

A. Wolman Pebble Count Protocol

1. **Grid Setup**: Delineate a transect across the bankfull width of the channel, starting at the water's edge.
2. **Blind Selection**: Walk across the transect. At every step, look away and touch the stream bed at the tip of your wading boot. Pick up the first particle your index finger contacts.
3. **Measurement**: Measure the **intermediate axis (B-axis)** of the particle using a standard gravelometer (in millimeters).
4. **Tally**: Log the particle size into the standard size categories (e.g., Silt/Clay *<0.062\text{ mm}*, Sand $0.062\text{--}2\text{ mm}*, Gravel $2\text{--}64\text{ mm}*, Cobble $64\text{--}256\text{ mm}*, Boulder* >256\text{ mm}$).
5. **Sample Size**: Repeat until you have collected exactly **100 random particles** across the reach cross-sections.

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4. Physical Habitat Parameter Rating

You will fill out the EPA RBP High-Gradient Physical Habitat Assessment Sheet at every station. Score the following ten parameters from **0 (Poor)** to **20 (Optimal)** to track physical habitat lifting:

Habitat Parameter	Scoring Range	Ecological Description & Field Indicator
1. Epifaunal Substrate	0 to 20	Relative abundance of natural cobble, gravel, and woody logs. Spawning and colonization shelter.

Habitat Parameter	Scoring Range	Ecological Description & Field Indicator
2. Embeddedness	0 to 20	The degree to which gravel and cobble are buried in fine sand/silt. Excellent is *<25\%* embedded.
3. Velocity-Depth Regimes	0 to 20	Presence of four patterns: Slow-Deep (pools), Slow-Shallow (runs), Fast-Deep (glides), Fast-Shallow (riffles).
4. Sediment Deposition	0 to 20	Formation of mid-channel sand bars or aggradation. Low score indicates high silt choking.
5. Channel Flow Status	0 to 20	Degree to which the channel is filled with water. Optimal reaches fill the bankfull bank margins.
6. Channel Alteration	0 to 20	Human impacts (culverts, concrete, bridges). Our restored reaches count as positive geomorphic lifting.
7. Frequency of Riffles	0 to 20	Ratio of distance between riffles divided by stream width. Stable streams feature riffles every 5 to 7 stream widths.
8. Bank Stability	0 to 20	Degree of bank erosion, slumping, and active clay failure. Restoration areas target high score.
9. Vegetative Protection	0 to 20	Riparian native vegetative cover (stems, roots, leaves). Protects banks from storm shear stress.
10. Riparian Zone Width	0 to 20	Width of native canopy buffer. Optimal is *>100\text{ feet}* of continuous undisturbed forest buffer.

5. Field Safety & Environmental Ethics

- **Decontamination Protocol**: To prevent the spread of invasive **Didymo (rock snot)** algae and the deadly fungal pathogen **Chytrid** (which decimates native salamanders), you must submerge all waders, boots, kick-nets, and D-nets in a **10% bleach solution** or **Virkon S disinfectant** for exactly 15 minutes between different mountain watersheds.
- **Wading Safety**: Never enter a stream reach if the stage height exceeds your waist or if water velocities exceed 3.5 ft/s. Always wade with a companion and use a wading staff to feel for deep scoured holes.